

A RGB-D geometry

RealSense D435i
Fixed camera; RGB + depth

Foreground + depth support
Task-object and robot masks

Table homography + tracker
Selected disc: centre c , radius r

B Asynchronous VLM

FastVLM-0.5B on laptop
Count visible human hands

Parsed count: 0 = OBJECT
Count ≥ 1 = HUMAN

Atomic frame-verdict bundle
Frame ID; proxy; inference times

C AEGIS: evidence age, score and margin

Evidence-linked score

$$\tau = t - \tilde{t}_{\text{cap}}$$

$$h_k = (1 - \eta)h_{k-1} + \eta y_k$$

$$p = P_0 + (h_k - P_0)e^{-\max(0, \tau)/T_d}$$

$$P_0 = 0.65, \eta = 0.95, T_d = 0.40 \text{ s}$$

$y_k = 1$ for HUMAN; 0 for OBJECT

Present-hazard intervention

$$m = \beta + p(R - d)$$

$$R = 60, d = 12, \beta = 12 \text{ mm}$$

AEGIS hold when $p \geq 0.50$

Blob jump > 50 mm: reset anchor;
cached semantic verdict is retained

No detected hazard: score/margin = 0

Proxy = host request time - server-reported frame age

D Shared carry controller: motion or hold

Shared hold checks

Absent / invalid evidence or HUMAN
Geometric frame age > 0.6 s
Confirmed depth-only / blocked goal

Otherwise: object-labelled route

Disc $r + m$; straight or cached detour
Planner: +35/70 mm; +18 mm padding
Per-step check: permit or HOLD

HOLD: no next carry increment

If permitted: jog / pose command

Log: frame + evidence, age, score, command-state and decision

Monitored carry only; command-state is not a measured TCP trajectory.